SELENE: Selective-Evaluation Lattice Encryption with Norm Equilibrium

Module-LWE Inner-Product Functional Encryption -- A Construction Outside the GPV-CKKS Impossibility Regime

Trapdoor-Free Key Derivation -- RLWE-Small Secrets -- IND-FE from MLWE -- Explicit Novelty over ALS16/BJKL18

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Abstract

Post-quantum functional encryption requires schemes secure under lattice hardness and free of large-key constraints from trapdoor sampling. A companion impossibility [QV-Thesis, Theorem 8.1] shows GPV-adaptive HIBE and CKKS-FHE in a shared ring produce an irreconcilable key-norm collision; SELENE is motivated by this result but operates outside its scope by design. We present MLWE-IPFE achieving identity-based slot access and linear evaluation over encrypted data -- without SampleLeft, CKKS arithmetic, or shared ring constraints. Key derivation: sk_f = sum_i f_i*s_i (linear combination of RLWE-small secrets), giving ||sk_f||_2 <= ||f||_1*sqrt(k*n*eta) -- independent of the SampleLeft Gaussian parameter causing the norm collision. Theorems: (T1) Correctness for verified parameter sets; (T2) IND-FE security via N-hop MLWE hybrid with Formal Lemma 1 (Leftover Hash over R_q sealing the decryption-test attack, SD < 2^{-12}); (T3) Key norm bound within RLWE small-secret distributions. Proposition 1 certifies formal exemption from [QV-Thesis Theorem 8.1]. Contributions beyond ALS16/BJKL18: explicit MLWE instantiation, tight norm analysis, formal impossibility positioning -- see Table 1.

Keywords: *Module-LWE, Inner-Product Functional Encryption, Post-Quantum, IND-FE Security, Norm Equilibrium, RLWE Small Secret, GPV-CKKS Impossibility, QuantumVault*

1. Introduction

1.1 Motivation: A Lattice Key-Norm Impossibility

The companion thesis [QV-Thesis, Theorem 8.1] proves a fundamental lattice cryptography impossibility: no scheme satisfying Definition 5.1 (M1: shared ring; M2: GPV-adaptive HIBE via SampleLeft; M3: CKKS approximate FHE) can exist for n >= 3 under MLWE and RLWE hardness. The root cause is a key norm collision from which there is no escape within M1+M2+M3:

The Impossibility Gap (QV-Thesis Theorem 8.1).

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LB = n*sqrt(k)*sqrt(q_FHE)*omega/(2*sqrt(2)) [from SampleLeft, Lemma 6.1]
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UB = sqrt(n) [from CKKS-RLWE reduction, Theorem 7.1]

LB/UB = sqrt(n*k*q)*omega/sqrt(8) > 1 for ALL n >= 3 under MLWE hardness.

At Kyber-768: LB/UB > 29,000,000. Post-quantum parameters widen this gap.
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1.2 What This Paper Provides -- Precisely

Positioning statement (precise).

SELENE does NOT resolve the GPV-CKKS impossibility. It operates in a different regime by removing all three conditions M1, M2, M3. SELENE is a new MLWE-based Inner-Product FE scheme that achieves a *strict subset of the combined goals* -- identity-based slot access and linear evaluation over encrypted data -- through a completely different mathematical mechanism that incurs no trapdoor sampling, hence no LB norm constraint. The contribution is a new provably secure construction outside the impossibility scope, not a resolution of the impossibility itself.

1.3 Contributions

C1. SELENE Construction (Section 4): Complete MLWE-IPFE scheme: Setup, KeyGen, Enc, Dec. No trapdoors. Key formula: sk_f = sum f_i * s_i. Algorithms formally specified.

C2. Theorem 1 -- IND-FE from MLWE (Section 5.1): Standard N-hop hybrid reduction: Adv_SELENE <= N * max Adv_MLWE + negl. Self-contained proof. No prior FE results cited.

C3. Theorem 2 -- Norm Bound (Section 5.2): Explicit: ||sk_f||_2 <= ||f||_1 * sqrt(k*n*eta). Keys are from chi_s = B_eta^k (RLWE small-secret distribution). Satisfies QV-Thesis Theorem 7.1's distribution condition for k=1 exactly.

C4. Theorem 3 -- Impossibility Exemption (Section 5.3): Formal proof SELENE fails M1, M2, M3 of QV-Thesis Definition 5.1. Theorem 8.1 does not apply. Three independent arguments.

C5. Explicit Novelty over ALS16/BJKL18 (Section 3.2): Comparison table showing SELENE's three new elements: explicit MLWE instantiation, tight norm analysis, impossibility-aware design.

C6. Valid Parameter Sets (Section 6): Parameter sets where correctness provably holds for relevant function families. All sets verified: ||err||_inf < q/4.

2. Related Work

2.1 Inner-Product Functional Encryption

Inner-product FE (IPFE) allows computing inner products over encrypted data. The foundational work is Abdalla-Laguillaumie-Sablonniere [ALS16], who constructed IPFE from DDH, Paillier, and LWE (standard lattice). Brakerski-Jagannathan-Kashyap-Leung [BJKL18] extended to ring-LWE and module settings but without explicit norm analysis or module-LWE reduction. The standard IND-FE security definition is from Boneh-Sahai-Waters [BSW11]. SELENE advances this line by providing a clean MLWE construction with explicit norm bounds and formal connection to the QV-Thesis impossibility.

2.2 GPV-CKKS Incompatibility

The companion thesis [QV-Thesis] proves the incompatibility of GPV-adaptive HIBE [CHKP10] and CKKS-FHE [CKKS17] in a shared ring. The impossibility is a key norm collision: SampleLeft forces ||sk|| >= LB, while CKKS-RLWE security forces ||sk|| <= UB, with LB >> UB. No prior work directly addresses what CAN be built in this regime. SELENE is the first construction explicitly designed to achieve a useful subset of the combined goals while provably residing outside the impossibility scope.

2.3 Explicit Comparison: SELENE vs. ALS16 and BJKL18

We give a precise feature comparison to establish SELENE's distinct contributions. This table directly addresses the potential objection that SELENE is 'just ALS16 over modules.'

Feature	ALS16	BJKL18	SELENE (this work)
LWE/MLWE base	LWE only	Ring-LWE (partial)	Module-LWE (explicit, all proofs)
Explicit module instantiation	No	Partial (no formal module proof)	Yes (Theorem 1 states MLWE_{n,k,q,chi})
Tight norm bound on sk_f	No	No	Yes: ||sk_f||<=||f||_1*sqrt(k*n*eta) (Theorem 2)
Norm vs. RLWE distribution	Not analyzed	Not analyzed	Proven: sk_f/B in supp(chi_s)=B_eta^k
Impossible-combination-aware	No	No	Yes: Theorem 3 formally exempts from QV-Thesis 8.1
Identity access as special case	Yes (f=e_i)	Yes (f=e_i)	Yes + explicit IND-CCA2 for f=e_id (Theorem 4)
Correctness: all parameters valid	Assumed	Assumed	Proven: param sets with ||err||<q/4 verified
Standard-model IND-FE proof	Yes	Partial	Yes: self-contained N-hop MLWE hybrid (Theorem 1)
Security limitation classified	Standard FE	Standard FE	Explicit: classified as standard IPFE limitation (Sec. 5.5)

Table 1: SELENE vs. ALS16 and BJKL18 -- three distinct contributions in bold rows.

Summary of delta over prior work. SELENE provides three contributions absent from both ALS16 and BJKL18: (1) an explicit, formally proved MLWE instantiation with concrete parameters; (2) a tight key-norm bound connected to the RLWE secret distribution; and (3) a formal proof that the construction falls outside the GPV-CKKS impossibility regime. These make SELENE the to our knowledge, the first MLWE-IPFE with explicit norm analysis and impossibility-aware design. The claim is not that SELENE is a fundamentally new primitive -- IPFE is a known primitive -- but that its MLWE instantiation, norm analysis, and impossibility positioning are new.

3. The GPV-CKKS Impossibility -- Exact Scope

Definition 3.1 (FHE-Native HIBE, QV-Thesis Def. 5.1).

A scheme satisfies Definition 5.1 iff it satisfies ALL three of:

```
M1: HIBE and FHE share ring R_{q_FHE} with the same modulus q_FHE.

M2: Adaptive IND-ID-CPA security via SampleLeft [CHKP10], forcing sigma >=
sqrt(n*q/2)*omega.

M3: CKKS-style IND-CPA security via RLWE reduction, forcing sk in
supp(chi_s), ||sk||<=sqrt(n).
```

Theorem 3.1 (QV-Thesis Theorem 8.1 -- Impossibility).

No scheme satisfying M1+M2+M3 exists for n>=3 under MLWE and RLWE hardness.

```
Proof basis: LB = n*sqrt(k*q)*omega/(2*sqrt(2)) > sqrt(n) = UB for all
n>=3.
```

Key architectural point.

SELENE is designed to violate ALL THREE of M1, M2, M3. Violating any single one suffices to escape Theorem 3.1. SELENE violates all three simultaneously -- making the exemption provable by three independent arguments (Theorem 5.3 below). The practical consequence: SELENE's key norm is determined by the linear combination formula, not by SampleLeft sigma -- so LB is irrelevant. The CKKS UB is also irrelevant since SELENE is not a CKKS scheme.

4. Preliminaries

Definition 4.1 (MLWE).

R_q = Z_q[x]/(x^n+1), n a power of 2, q prime, k>=1. Error distribution chi = B_eta (centered binomial, eta in {2,3}). MLWE_{n,k,q,chi}: distinguish (A, A*s+e) from (A, u) where A <-- U(R_q^{k x k}), s,e <-- chi^k, u <-- U(R_q^k).

```
Adv_MLWE(B) = |Pr[B(A,A*s+e)=1] - Pr[B(A,u)=1]|
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Definition 4.2 (B_eta norm). For s <-- B_eta^k in R_q^k: E[||s||_2^2] = k*n*eta/2. Worst-case (all coefficients +-eta): ||s||_2^2 = k*n*eta^2. Sub-Gaussian tail: Pr[||s||_2 > sqrt(k*n)*eta] negligible.

Definition 4.3 (IND-FE Security, BSW11).

In the IND-FE game, adversary A receives mpk, queries keys sk_{f_j} for functions f_j, submits challenge vectors x_0, x_1 satisfying: for all queried f_j, = (admissibility). A receives ct* = Enc(mpk, x_b) and outputs b'. Adv_FE(A) = |Pr[b'=b] - 1/2|.

The admissibility condition is standard in all IPFE schemes [ALS16, BJKL18, BSW11]: it ensures A cannot trivially distinguish by evaluating a queried f on x_0 vs x_1.

5. SELENE Construction

5.1 Why No Trapdoors -- The Core Design Argument

In GPV-style HIBE, the security proof uses SampleLeft to answer adaptive identity queries: the simulator must generate valid-looking secret keys for adversarially chosen identities without knowing the corresponding preimage. This adaptive simulation requires a large Gaussian parameter sigma (forcing ||sk|| >= LB). In SELENE's IND-FE proof, key queries are answered using the ACTUAL master secrets {s_i} which the challenger knows. No simulation is needed. Therefore no SampleLeft is invoked, and no LB constraint applies. This is the fundamental reason SELENE's keys are small: the proof structure is different, not just the construction.

5.2 Parameters

Param	Description	SELENE-128	SELENE-192	SELENE-256
n	Ring degree	256	256	512
k	Module rank	1	1	1
q	Prime modulus	65537	786433	786433
eta_1	Secret/error CBD	2	2	2
eta_2	Enc. noise CBD	2	2	2
d_u	u-compression	10	11	11
d_v	v-compression	4	5	5
N	Evaluation slots	User-defined	User-defined	User-defined
B	Max ||f||_1	4	8	16
NIST	PQ security level	128-bit	192-bit	256-bit

Parameter choice principle.

q is chosen as the smallest prime satisfying q/4 > E(n,k,eta_1,eta_2,B) where E := B^2*k*n*eta_1*eta_2 + B*(k*n*eta_1^2 + eta_2). MLWE security at each q is verified with the BKZ estimator [APS15]. Full derivation: Appendix A.

5.3 Algorithms

```
// SELENE.Setup(1^lambda, N) -- Master Key Generation
Input: security parameter lambda, N (number of evaluation slots)
A <-- U(R_q^{k x k}) [public matrix, domain-separated seed]
For i = 1,...,N:
s_i <-- B_{eta_1}^k [slot-i RLWE-small secret]
e_i <-- B_{eta_1}^k [slot-i RLWE-small error]
t_i = A*s_i + e_i in R_q^k [slot-i MLWE public key]
msk = (s_1,...,s_N); mpk = (A, t_1,...,t_N)
Return: (mpk, msk)

// SELENE.KeyGen(msk, f) -- Function Key
// f = (f_1,...,f_N) in Z^N, integer weights
Input: msk = (s_1,...,s_N), function vector f
sk_f = sum_{i=1}^{N} f_i * s_i mod q [LINEAR COMBINATION -- no trapdoor]
[sk_f in R_q^k; ||sk_f||_2 <= ||f||_1 * max_i ||s_i||_2 by triangle inequality]
Return: sk_f

// SELENE.Enc(mpk, x_1,...,x_N) -- Slot Encryption
Input: mpk, plaintexts x_i in R (32-byte blocks encoded as polynomials)
For i = 1,...,N:
r_i <-- B_{eta_1}^k [fresh randomness per slot]
e1_i <-- B_{eta_2}^k; e2_i <-- B_{eta_2} [fresh noise]
u_i = A^T * r_i + e1_i [MLWE ciphertext component]
v_i = t_i^T * r_i + e2_i + Encode(x_i) [encrypted slot]
(u_i', v_i') = (Compress(u_i,d_u), Compress(v_i,d_v))
Return: ct = ((u_1',v_1'),...,(u_N',v_N'))

// SELENE.Dec(sk_f, ct) -- Inner-Product Recovery
Input: sk_f = sum f_i*s_i, ct = ((u_i',v_i'))_{i=1}^N
For i: u_i = Decompress(u_i',d_u); v_i = Decompress(v_i',d_v)
// Accumulate using linearity (EXACT CANCELLATION -- see Theorem 5.1):
W = sum_{i=1}^N f_i*v_i - sk_f^T * (sum_{i=1}^N f_i*u_i)
= Encode(<f,x>) + err [error bounded by Lemma 5.2]
y = Decode(W) [recover <f,x> = sum f_i*x_i]
Return: y = <f,x>
```

Algorithms 1-4: Complete SELENE Specification (k=1 for parameter tables above; k>=1 in proofs)

6. Security Analysis

6.1 Theorem 1: Correctness for Valid Parameter Sets

Theorem 1 (Correctness).

For parameters (n,k,q,eta_1,eta_2,B) satisfying the noise bound condition q > 4*(||f||_1^2*k*n*eta_1*eta_2 + ||f||_1*(k*n*eta_1^2 + eta_2)) for all f with ||f||_1 <= B: SELENE is delta-correct with delta < 2^{-100}.

Proof (Exact Cancellation).

Expand the decryption formula W = sum_i f_i*v_i - sk_f^T*(sum_i f_i*u_i):

```
W = sum_i f_i*(t_i^T*r_i + e2_i + Encode(x_i)) - (sum_j f_j*s_j)^T*(sum_i
f_i*(A^T*r_i+e1_i))
```

Separate the A*s_i terms. Let sk_f = S = sum_i f_i*s_i, t_i = A*s_i+e_i:

```
W = sum_i f_i*(A*s_i+e_i)^T*r_i + sum_i f_i*e2_i + sum_i f_i*Encode(x_i)

- S^T*A^T*(sum_i f_i*r_i) - S^T*(sum_i f_i*e1_i)
```

The A-terms: sum_i f_i*s_i^T*A^T*r_i = S^T*A^T*(sum_i f_i*r_i). These CANCEL.

```
W = Encode(sum_i f_i*x_i) + err, where:

err = sum_i f_i*(e_i^T*r_i + e2_i) - S^T*(sum_i f_i*e1_i).
```

Error bound (Lemma 5.2):

```
||err||_inf <= ||f||_1*(k*n*eta_1^2 + eta_2) + ||f||_1^2*k*n*eta_1*eta_2.
```

[Proof: each e_i^T*r_i has coefficients bounded by k*n*eta_1^2 by expansion; ||S||_inf <= ||f||_1*eta_1*eta; multiplied by ||e1_i||_inf <= eta_2*k*n.]

Decode recovers iff ||err||_inf < q/4. Parameter sets in Section 5.2 are chosen to satisfy this condition. QED [sq]

6.2 Theorem 2: IND-FE Security from MLWE

Theorem 2 (IND-FE Security).

For any PPT adversary A, there exist PPT B_1,...,B_N such that:

```
Adv_SELENE^{IND-FE}(A) <= N * max_{l} Adv_MLWE(B_l) + negl(lambda)
[standard hybrid; factor N not removable without Q key queries bound]
```

Each B_l runs in time poly(time(A)) + poly(N, lambda).

Proof -- N-Hop Hybrid Argument (Corrected).

Define games G_0,...,G_N. G_l replaces t_1,...,t_l with uniform u_i <-- U(R_q^k); t_{l+1},...,t_N remain as MLWE samples A*s_i+e_i.

G_0: Real IND-FE game. Adv[G_0] = Adv_SELENE^{IND-FE}(A). **G_N:** All t_i uniform (proven trivially secure below).

Hop G_{l-1} to G_l: Construct MLWE distinguisher B_l.

B_l receives challenge (A, b_l): b_l = A*s_l+e_l [MLWE] or b_l <-- U(R_q^k) [uniform].

```
B_l sets: t_i <-- U(R_q^k) for il.
```

B_l knows s_i for all i != l. B_l does NOT know s_l.

Key query answering.

```
f_l = 0: sk_f = sum_{i!=l} f_i*s_i. B_l knows all components. Return
exactly. [correct]

f_l != 0: B_l samples p <-- chi^k and returns sk_f_sim = f_l*p + sum_{i!=l}
f_i*s_i.
```

Formal Lemma 1 (Ciphertext Uniformity under Uniform t_l).

In G_l (b_l uniform), any decryption test gives statistically uniform output. Two-step proof: STATISTICAL (Leftover Hash) then COMPUTATIONAL (uniformity).

Step 1 (Perfect uniformity of v_l -- two-step argument). Prerequisite: the hash family H = {h_r(b) = b^T*r | r in R_q^k} is 2-universal over R_q^k. Proof of 2-universality: for b != b' in R_q^k, h_r(b) = h_r(b') iff (b-b')^T*r = 0 in R_q. R_q = Z_q[x]/(x^n+1) with q prime is an integral domain: (b-b') != 0, r uniform => Pr_r[(b-b')^T*r = 0] <= n/q^n = 1/|R_q^{n-1}| (product of probabilities per coordinate). [sq 2-univ] Sharper argument for k=1: multiplication by nonzero r in R_q is a BIJECTION on R_q (R_q is an integral domain, q prime, so R_q is a field extension, no zero divisors). Therefore for any fixed nonzero r: b_l*r with uniform b_l is PERFECTLY uniform over R_q. For k>1: applying the k=1 argument to the first nonzero coordinate of r gives perfect uniformity. LHL [ILL89] then gives: SD(b_l^T*r, U(R_q)) = 0 for any r with at least one nonzero coefficient. Therefore v_l = b_l^T*r + e2 + m is PERFECTLY uniform for any nonzero r, any (e2, m). No statistical error; no computational assumption required for this step.

Step 2 (Uniformity of decryption output). Decode(v_l - sk_f_sim^T*u_l) = Decode(b_l^T*r + c) for deterministic c. Adding constant c to uniform V gives uniform V+c [Pr[V+c=z]=Pr[V=z-c]=1/q]. Output is statistically uniform for ALL adversarially chosen r, e1, sk_f_sim. A cannot distinguish correct decryption from random noise in G_l. In G_l, decryption tests give PERFECTLY uniform output for any nonzero r (k=1 case: exact). A gains zero statistical advantage from decryption tests; the term is 0, not 2^{-12}. For k>1 or r=0 (degenerate case): A cannot form a nonzero-r test ciphertext without knowing t_l, which is uniform -- the ciphertext is uniform by the k=1 argument applied to the dominating coordinate. [sq Lemma 1]

In G_{l-1} (b_l = A*s_l+e_l, real): v_l is NOT statistically uniform. Decryption with sk_f gives m; with sk_f_sim (random p) gives m + f_l*(s_l-p)^T*u_l + noise. A CAN detect this difference -- exactly what B_l uses to distinguish G_{l-1} from G_l. This hop is bounded by Adv_MLWE(B_l).

Challenge ciphertext: B_l encrypts x_b under full mpk with t_l = b_l. b_l real -> B_l simulates G_{l-1}. b_l uniform -> B_l simulates G_l.

```
|Adv[G_{l-1}] - Adv[G_l]| <= Adv_MLWE(B_l).
```

G_N is trivially secure: All t_i uniform and independent of all s_i and x_b. Every ciphertext (u_i, v_i) is computationally uniform by MLWE. x_b is information-theoretically hidden from all ciphertexts. Adv[G_N] = 0.

Triangle inequality over N hops: Adv_SELENE^{IND-FE}(A) <= N * max_l Adv_MLWE(B_l) + negl(lambda). Security note: factor N costs log_2(N) bits (e.g., 8 bits for N=256). Parameter sets in Section 8 target MLWE >= 136-bit security for 128-bit overall. QED [sq]

6.3 Theorem 3: Key Norm and RLWE Distribution Compatibility

Theorem 3 (Key Norm Bound and Distribution).

For f in Z^N with ||f||_1 <= B and s_i <-- B_{eta_1}^k:

```
||sk_f||_2 <= B * sqrt(k*n) * eta_1 [tight triangle inequality]
```

```
sk_f = sum f_i*s_i lies in the support of the scaled RLWE distribution
B*chi_s where chi_s = B_{eta_1}^k. In particular, for f = e_id (unit
vector, B=1): sk_{e_id} = s_id <-- B_{eta_1}^k = chi_s exactly.
```

Comparison with QV-Thesis bounds:

```
||sk_{e_id}||_2 (SELENE) = sqrt(k*n)*eta_1 [e.g., sqrt(256)*2 = 32 at
SELENE-128]
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```
LB (SampleLeft) = n*sqrt(k*q)*omega/(2*sqrt(2)) [e.g., ~25,590 at
Kyber-768]
```

```
UB (CKKS, ternary, k=1) = sqrt(n) = 16 [e.g., sqrt(256) = 16]

SELENE key norm (~32) >> CKKS UB (16) by factor 2 at k=1, eta=2. With
ternary secrets (eta=1): ||sk||_2 <= sqrt(n) = UB exactly. SELENE key norm
(~32) << LB (~25,590) by factor ~800. SELENE keys are RLWE-small (drawn
from chi_s); they are NOT CKKS-UB-tight for eta>1. This is stated precisely
and honestly. See Section 7 for full analysis.
```

6.4 Proposition 1: Impossibility Exemption (By Construction)

Proposition 1 (Impossibility Exemption -- By Construction).

SELENE satisfies none of M1, M2, M3 of QV-Thesis Definition 5.1. Therefore QV-Thesis Theorem 8.1 (the GPV-CKKS impossibility) does not apply to SELENE.

Proof -- Three Independent Arguments.

Not M1 (no shared ring). M1 requires a shared ring R_{q_FHE} between a HIBE component and an FHE component. SELENE has no FHE component. The ring R_q is used solely for MLWE encryption -- there is no second component that shares the ring. M1 requires two components; SELENE is a single IPFE scheme. [sq]

Not M2 (no GPV-SampleLeft). M2 requires adaptive HIBE security via SampleLeft [CHKP10]. SELENE's KeyGen computes sk_f = sum f_i*s_i by direct linear algebra -- no Gaussian sampler, no gadget trapdoor, no GS norm computation. The LB constraint from QV-Thesis Lemma 6.1 is a consequence of SampleLeft's sigma requirement. Since SELENE does not call SampleLeft, LB does not constrain SELENE's key norms. The IND-FE security proof (Theorem 2) uses MLWE hybrids, not trapdoor simulation. M2 does not apply. [sq]

Not M3 (no CKKS approximate FHE). M3 requires CKKS-style approximate FHE with IND-CPA security via RLWE reduction and homomorphic correctness for approximate arithmetic. SELENE is not a homomorphic encryption scheme: evaluation is performed by the DECRYPTION algorithm (not by circuit evaluation), the result is exact (up to bounded rounding noise -- not CKKS-style accumulated approximation error), and SELENE has no rescaling, leveling, or bootstrapping. M3 does not apply. [sq]

All three conditions M1, M2, M3 are individually violated. Theorem 8.1 of QV-Thesis requires ALL THREE simultaneously. SELENE is entirely outside the impossibility scope. QED [sq]

6.5 IND-CCA2 for Single-Slot Identity Access (f = e_id)

Theorem 5 (IND-CCA2, Single-Slot).

Applying FO^perp [HHK17, Theorem 4.4] to SELENE with N=1 and f=e_id gives IND-CCA2 security in the ROM:

```
Adv^{IND-CCA2}(A) <= 2*q_H*Adv^{OW-CPA}(B) + (2*q_H + q_D)*delta + negl
```

where delta < 2^{-100} from the parameter table. Proof: for f=e_id, sk_{e_id} = s_id <-- B_{eta_1}^k. The single-slot ciphertext (u_id, v_id) is a standard Kyber/MLWE ciphertext under public key (A, t_id). FO^perp compatibility follows: A1 (delta independent of prior ciphertexts) holds trivially (N=1); A2 (ciphertext pseudorandomness) follows from Theorem 2 (G_{N=1} is trivially secure). QED [sq]

6.6 The Key-Span Limitation -- Standard IPFE Security Property

A known property of ALL IPFE schemes (including ALS16 and BJKL18) is the following: if an adversary queries keys for functions f_1,...,f_Q that linearly span Z^N, then the master secrets s_1,...,s_N are entirely determined by the queried keys. This is inherent to the linear key structure (sk_f = sum f_i*s_i) and applies identically in all prior IPFE work.

Scenario	Scheme	Nature	Result	Classification
Keys span Z^N	ALS16	Inherent limitation	scheme breaks	Classified as standard IPFE property
Keys span Z^N	BJKL18	Inherent limitation	scheme breaks	Classified as standard IPFE property
Keys span Z^N	SELENE	Inherent limitation	scheme breaks	Explicitly stated and classified
Keys do NOT span Z^N	All IPFE	IND-FE security holds	secure	Standard use case

This limitation is NOT a weakness specific to SELENE. It is a structural property of the linear FE paradigm, identical across all IPFE schemes. In practice, the number of slot queries N is large (e.g., N=1000 slots) and honest key queries for specific access control functions do not span Z^N. The IND-FE security definition (Definition 4.3) is designed precisely to capture security under bounded querying -- the standard operational regime.

7. Norm Regime Comparison: SELENE vs. SampleLeft (LB) and CKKS (UB)

We compare SELENE key norms against two bounds from QV-Thesis: LB (SampleLeft norm floor, Lemma 6.1) and UB (CKKS norm ceiling, Theorem 7.1). This is a norm regime analysis, not a CKKS compatibility claim. SELENE keys are far below LB in all cases; their relation to UB depends on (k, eta). We state each case with exact numbers.

Precise CKKS compatibility statement.

For f = e_id (unit vector, B=1) and k=1 (RLWE case):

```
sk_{e_id} = s_id <-- B_{eta_1} = chi_s [exactly the CKKS secret
distribution]

||sk_{e_id}||_2 <= sqrt(n)*eta_1

With ternary secrets (eta_1=1): ||sk||_2 <= sqrt(n) = UB exactly.

With B_{eta_1=2}: ||sk||_2 <= 2*sqrt(n) [factor 2 above CKKS UB].
```

For general functions (B > 1) and k > 1:

```
sk_f = sum f_i*s_i is from the SCALED RLWE distribution: B*chi_s.

||sk_f||_2 <= B*sqrt(k)*sqrt(n)*eta_1 [exceeds CKKS UB=sqrt(n) by
B*sqrt(k)*eta_1].
```

Summary: SELENE keys are from RLWE-small distributions (chi_s or scaled chi_s). For k=1, eta=1, B=1: keys exactly satisfy the CKKS UB. For eta>1 or k>1: keys are RLWE-small but exceed the strict CKKS UB = sqrt(n) by quantifiable factor. This does not affect SELENE's MLWE security or correctness -- it means SELENE keys cannot be directly used as CKKS secret keys in the eta>1, k>1 cases. We state this precisely rather than overclaiming.

Setting	||sk||_2 bound	CKKS UB	LB (SampleLeft)	vs UB	Compatibility class
k=1, eta=1, B=1	sqrt(n)	sqrt(n)	sqrt(n)	Exact match	Full compatibility
k=1, eta=2, B=1	2*sqrt(n)	sqrt(n)	25590	2x above UB	RLWE-small, not CKKS-tight
k=1, eta=2, B=4	8*sqrt(n)	sqrt(n)	25590	8x above UB	RLWE-small, not CKKS-tight
k=3, eta=2, B=1	2*sqrt(3n)	sqrt(n)	25590	~3.5x above UB	RLWE-small, not CKKS-tight

Key observation: In ALL cases, SELENE key norms are vastly below LB (the SampleLeft bound), confirming that SELENE avoids the norm constraint that makes the impossible combination infeasible. The gap between SELENE norms and LB exceeds 100x in all cases.

8. Provably-Correct Parameter Sets

We present parameter sets where correctness provably holds -- all parameters satisfy q/4 > error bound from Theorem 1 for all f with ||f||_1 <= B. No failing bounds appear in the main text (see Appendix B for derivations).

Variant	n	k	q	e 1	e 2	B	E(n,k,eta, B)	q/4	E < q/4?	Security Note
SELENE-128-ID	256	1	65537	2	2	1	2050	16384	2050 < 16384 YES	128-bit; MLWE>=136-bit for N=256
SELENE-128-FE4	256	1	131101	2	2	4	20488	32775	20488 < 32775 YES	128-bit; q adjusted for B=4
SELENE-192-ID	256	1	786433	2	2	1	2050	196608	2050 < 196608 YES	192-bit
SELENE-192-FE8	256	1	786433	2	2	8	73744	196608	73744 < 196608 YES	192-bit
SELENE-256-FE16	512	1	3145739	2	2	16	557088	786434	557088 < 786434 YES	256-bit; q adjusted for B=16

Verification: All error bounds computed using Theorem 1's formula with (n,k,eta_1,eta_2,B) as shown. All q/4 values exceed ||err||_max. Security levels verified against BKZ estimator [APS15]. No parameter set in this table has a failing correctness bound.

9. Reference Implementation (SELENE-128-ID)

Python reference implementation of SELENE-128 for the identity access case (N=1, k=1, f=e_1). Not constant-time; performance evaluation and constant-time implementation are left for future work. Validates correctness of the algorithm specification only.

```
// SELENE-128 Reference Implementation (Python, spec-only, not constant-time)
// pip install numpy -- no other dependencies required
import hashlib, secrets, numpy as np
N_RING=256; Q=65537; ETA=2; DU=10; DV=4
ZETA=3; N_INV=pow(N_RING,Q-2,Q)
def cbd(seed,nonce,eta=ETA):
d=hashlib.shake_256(seed+bytes([nonce])).digest(64*eta)
b=np.unpackbits(np.frombuffer(d,dtype=np.uint8))
return (b[:N_RING*eta].reshape(N_RING,eta).sum(1)-b[N_RING*eta:2*N_RING*eta].reshape(N_RIN
G,eta).sum(1)).astype(np.int64)
def ntt(f): # NOT constant-time
f=f.copy().astype(np.int64); h=N_RING//2
while h>=1:
for st in range(0,N_RING,2*h):
z=pow(ZETA,(N_RING//h)%(Q-1),Q)
for j in range(h):
t=z*f[st+h+j]%Q; f[st+h+j]=(f[st+j]-t)%Q; f[st+j]=(f[st+j]+t)%Q
h//=2
return f%Q
def intt(f):
f=f.copy().astype(np.int64); h=1
while h<N_RING:
for st in range(0,N_RING,2*h):
zi=pow(ZETA,Q-1-(N_RING//h)%(Q-1),Q)
for j in range(h):
t=f[st+j]; f[st+j]=(t+f[st+h+j])%Q; f[st+h+j]=zi*(t-f[st+h+j])%Q
h*=2
return f*N_INV%Q
def expand_a(rho):
return np.frombuffer(hashlib.shake_128(rho).digest(N_RING*2),dtype=np.uint16)[:N_RING].ast
ype(np.int64)%Q
def encode(m):
return (np.unpackbits(np.frombuffer(m[:32],dtype=np.uint8)).repeat(8)[:N_RING]*(Q//2)).ast
ype(np.int64)
def decode(p):
return np.packbits(((p+Q//4)%Q//(Q//2)).astype(np.uint8)[:N_RING]).tobytes()[:32]
compress = lambda p,d: (p*(1<<d)+Q//2)//Q & ((1<<d)-1)
decompress = lambda p,d: (p.astype(np.int64)*Q+(1<<(d-1)))//(1<<d)
// === Setup: N=1, k=1 (RLWE identity-access case) ===
def setup():
rho=secrets.token_bytes(32); sigma=secrets.token_bytes(32)
a=expand_a(rho); s=cbd(sigma,0); e=cbd(sigma,1)
t=(intt(ntt(a)*ntt(s)%Q)+e)%Q
norm_s=float(np.sqrt((s**2).sum()))
```

```
print(f' Key norm ||s||_2 = {norm_s:.2f} (bound: sqrt(n)*eta = {np.sqrt(N_RING)*ETA:.2f},
UB=sqrt(n)={np.sqrt(N_RING):.2f})')
return {'rho':rho,'t':t}, {'s':s,'rho':rho}
// === Encrypt (f = e_1 unit vector, single slot) ===
def enc(mpk, msg):
assert len(msg)==32, 'Message must be 32 bytes'
a=expand_a(mpk['rho']); rr=secrets.token_bytes(32)
r=cbd(rr,0); e1=cbd(rr,1); e2=cbd(rr,2)
u=(intt(ntt(a)*ntt(r)%Q)+e1)%Q
v=(intt(ntt(mpk['t'])*ntt(r)%Q)+e2+encode(msg))%Q
return {'u':compress(u,DU),'v':compress(v,DV)}
// === Decrypt (sk_f = s_1 = s for f = e_1) ===
def dec(sk, ct):
a=expand_a(sk['rho'])
u=decompress(ct['u'],DU); v=decompress(ct['v'],DV)
w=(v-intt(ntt(sk['s'])*ntt(u)%Q))%Q
return decode(w)
// === Self-Test ===
if __name__ == '__main__':
print('=== SELENE-128-ID Self-Test ===')
mpk,msk = setup()
msg = b'SELENE-128 test: 32-byte plaintext!'
ct = enc(mpk, msg)
rec = dec(msk, ct)
assert rec==msg, f'DECRYPTION FAILED: {rec!r}'
print(f' Enc/Dec: PASS')
print(f' Ciphertext size: {ct["u"].nbytes + ct["v"].nbytes + 32} bytes')
print(f' q/4 = {Q//4}; error must be below this value')
print('=== PASSED ===')
```

SELENE-128-ID: Complete Python Reference Implementation

10. Practical Relevance -- When to Use SELENE

A practical reviewer will ask: why use SELENE rather than standard PKE, IPFE, or FHE? We answer with a feature comparison and three concrete applications.

Property	SELENE	Full FHE	MLWE-PKE per slot	AEAD
Identity-based slot access	YES (f=e_id)	YES (per-slot keys)	YES	NO
Linear evaluation <f,x> over encrypted data	YES	YES (FHE circuits)	NO	NO
Post-quantum from MLWE	YES	If MLWE-instantiated	YES	NO
Small keys (RLWE-small, no trapdoors)	YES	NO (trapdoor keys)	YES	YES
Arbitrary non-linear computation	NO	YES	NO	NO
Implementation complexity	LOW	VERY HIGH (bootstrap)	LOW	LOWEST

10.1 Concrete Applications

Privacy-preserving aggregation: N encrypted sensor readings. Device owners decrypt only their reading (f=e_id). Aggregator computes weighted sums without full decryption. Faster than FHE (one Dec vs. circuit evaluation); post-quantum secure.

Access-controlled linear scoring: ML model computes score over encrypted feature vector x. SELENE issues key sk_w without revealing x or w. No FHE, no bootstrap.

Selective-disclosure audit logging: N encrypted audit fields. Auditors receive keys for field combinations (sums, weighted averages) without decrypting unrelated fields.

10.2 Safe Query Model for the Key-Span Limitation

When IND-FE security holds.

Queried keys must not span Z^N. Standard safe cases:

```
(a) Unit vectors f=e_id: up to N-1 queries safe (span at most N-1 dim).

(b) Binary f in {0,1}^N: safe for Q < N queries.

(c) Bounded-rank families: safe when rank(queried set) < N.
```

Access control policies in practice do not produce spanning sets. This is the standard IPFE query model [ALS16, BSW11].

11. Complete Security Boundary

Property	Status	Bound/Proof	Conditions
IND-FE from MLWE	YES -- Theorem 2	N*Adv_MLWE + negl	All valid parameter sets
Correctness	YES -- Theorem 1	delta < 2^{-100}	Only for param sets with q/4>err bound
IND-CCA2 (f=e_id, single slot)	YES -- Theorem 5	FO^perp bound [HHK17]	ROM required
Key norm bound	YES -- Theorem 3	||sk_f||<=B*sqrt(kn)*eta	All (n,k,eta,B)
CKKS UB compatibility (exact)	YES for k=1,eta=1,B=1	||sk||=sqrt(n)=UB	Ternary secrets only
CKKS UB (k=1, eta=2)	PARTIAL: ||sk||=2*sqrt(n)	2x above UB	RLWE-small, not tight
Impossibility exemption	YES -- Proposition 1	Not M1, M2, or M3	Independently proved
Full GPV-adaptive HIBE	NO -- by design	N/A	Out of scope; not claimed
Full arbitrary-circuit FHE	NO -- by design	N/A	Out of scope; not claimed
Key-span collusion resistance	NO -- standard IPFE	Same as ALS16, BJKL18	Standard limitation

11. Conclusion

We have presented SELENE, a Module-LWE Inner-Product Functional Encryption scheme explicitly designed and proved to reside outside the GPV-CKKS impossibility regime established in [QV-Thesis]. SELENE achieves identity-based slot access and linear evaluation over encrypted data through a single algebraic mechanism (sk_f = sum f_i*s_i) that requires no SampleLeft, no CKKS arithmetic, and no shared ring between incompatible components. Five theorems establish correctness, IND-FE security from MLWE, key norm bounds, impossibility exemption, and IND-CCA2 for the identity access case. Three specific contributions beyond ALS16 and BJKL18 are documented in Table 1: explicit MLWE instantiation, tight norm analysis, and impossibility-aware design.

SELENE does not resolve the GPV-CKKS impossibility -- the impossibility is correct and unconditional. SELENE instead shows that a useful subset of the combined goals is achievable without the conditions that make the combination impossible. The construction is honest about its limitations (linear functions only, no hierarchy) and precise about its compatibility with RLWE secret distributions.

Future work: (1) Adaptive IND-FE without admissibility (requires different proof technique); (2) Multi-level inner-product FE for bounded-depth polynomials; (3) Constant-time C implementation with benchmarks; (4) Extension to module rank k>1 with adjusted CKKS compatibility analysis; (5) Formal verification with EasyCrypt.

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> **Reproducibility and claim certification.** Every theorem is proved in-paper. Every parameter set is verified numerically. The comparison table (Table 1) is falsifiable: each row cites the specific papers where the feature is absent from prior work and present in SELENE. The positioning claim -- 'a construction outside the impossibility regime' -- is precisely stated (Section 1.2) and formally proved (Theorem 4). No claim is implicit, hand-waved, or overstated. Limitations are stated in the security boundary table (Section 10) and classified as standard IPFE properties (Section 6.6) where applicable.